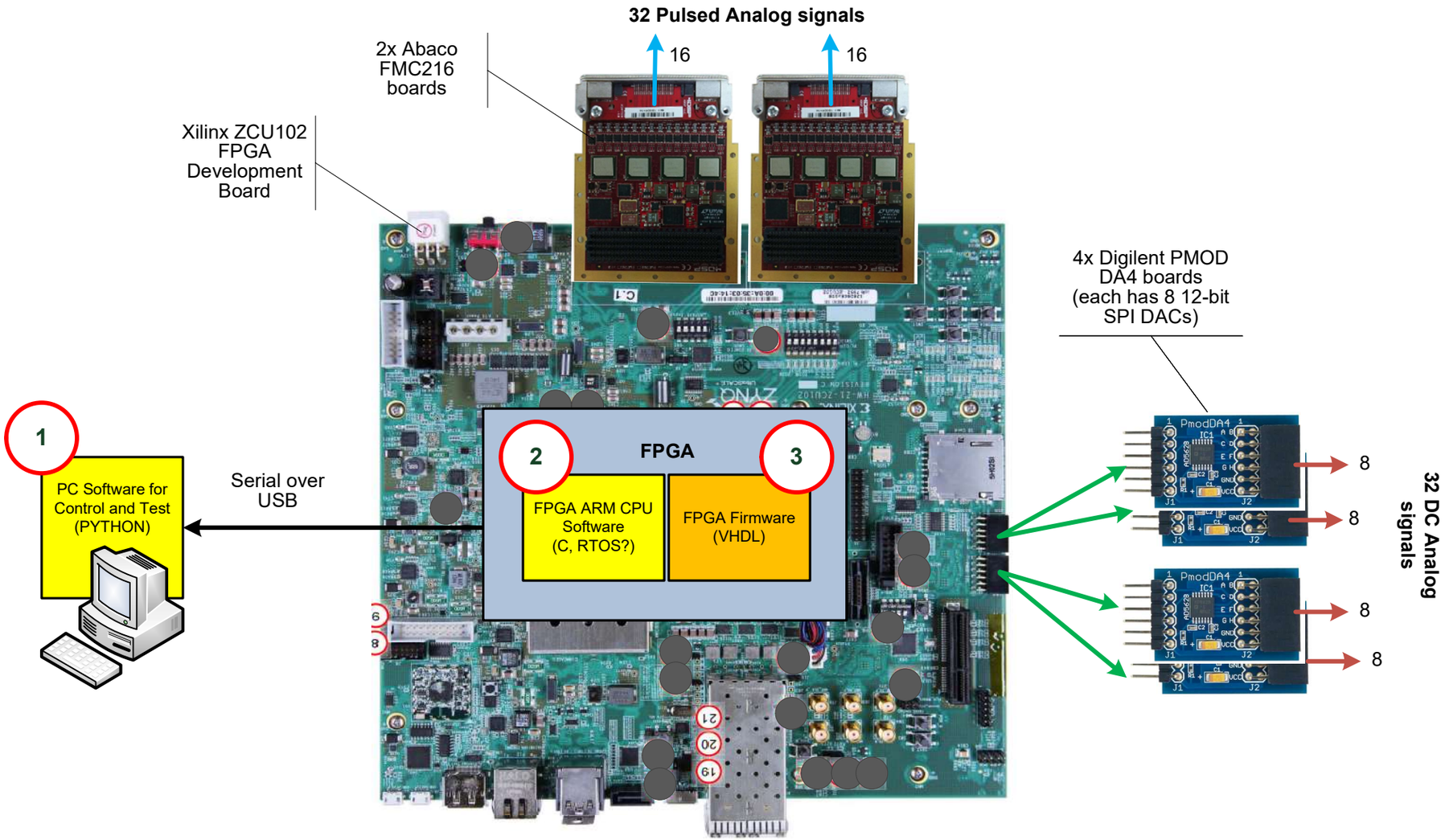
Laser Control Project Components



1. PC Software for Control and Test (PYTHON, Tk for GUI) Communicates with FPGA ARM CPU over 115,200 bps serial link.
2. FPGA ARM CPU (PS) Software (C, RTOS?) Developed in Vitis environment. Controls and monitors JESD blocks, Interprets commands from PC interface. Programs PL RAM tables etc.
3. FPGA Firmware. (VHDL/SystemVerilog) Custom logic connected to PS through AXI-lite interface. Contains DC inteface channels, Pulse generators and JESD IP blocks.